Math 220B HW 2

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Problem 1

Let R be a semilocal ring and let M and N be two R-modules such that $M \otimes_R N = 0$. Prove or disprove that either $M = 0$ or $N = 0$.

Solution: We'll disprove the statement by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, which is semilocal ring because it only has finite maximal ideals.
Now, consider the canonical projection $\pi_2 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\pi_3 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$, notice that since $6\mathbb{Z} = 2\mathbb{Z} \cap 3\mathbb{Z}$, one has the ideal $6\mathbb{Z} \subset \ker(\pi_2) = 2\mathbb{Z}$ and $6\mathbb{Z} \subset \ker(\pi_3) = 3\mathbb{Z}$. Hence, by Generalized First Isomorphism Theorem, with the canonical projection $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, the above two maps uniquely factor into surjective maps $\varphi_2 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\varphi_3 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ as follow:

$$\begin{tikzcd}
\mathbb{Z} \arrow[rr, "\pi_2"] \arrow[dr, two heads, "\pi_6"'] & & \mathbb{Z}/2\mathbb{Z} \\
& \mathbb{Z}/6\mathbb{Z} \arrow[ur, two heads, "\varphi_2"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbb{Z} \arrow[rr, "\pi_3"] \arrow[dr, two heads, "\pi_6"'] & & \mathbb{Z}/3\mathbb{Z} \\
& \mathbb{Z}/6\mathbb{Z} \arrow[ur, two heads, "\varphi_3"'] &
\end{tikzcd}$$

Hence, $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}$ can be realized as R-module, via the map φ_2 and φ_3 (in particular, one has $\varphi_2(\overline{1}_6) = \overline{1}_2$ and $\varphi_3(\overline{1}_6) = \overline{1}_3$ as the map, based on the above commutative diagrams).
Now, consider the tensor $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z}$, we claim that it is 0: Recall that $2, 3 \in \mathbb{Z}$ are coprime, with Bezout's Lemma there exists integers $s, t \in \mathbb{Z}$, such that $1 = 2s + 3t$. Then, under the projection map $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, one knows there exists $\overline{s}_6, \overline{t}_6 \in \mathbb{Z}/6\mathbb{Z}$, such that the following holds:

$$\overline{1}_6 = \overline{2}_6 \cdot \overline{s}_6 + \overline{3}_6 \cdot \overline{t}_6 \tag{1.1}$$

Then, for any $\overline{a}_2 \in \mathbb{Z}/2\mathbb{Z}$ and $\overline{b}_3 \in \mathbb{Z}/3\mathbb{Z}$, one has the following:

$$\begin{aligned}
\overline{a}_2 \otimes \overline{b}_3 &= \overline{1}_6 \cdot \left(\overline{a}_2 \otimes \overline{b}_3\right) = \left(\overline{2}_6 \cdot \overline{s}_6 + \overline{3}_6 \cdot \overline{t}_6\right) \cdot \left(\overline{a}_2 \otimes \overline{b}_3\right) \\
&= \overline{s}_6 \cdot \left(\left(\overline{2}_6 \cdot \overline{a}_2\right) \otimes \overline{b}_3\right) + \overline{t}_6 \cdot \left(\overline{a}_2 \otimes \left(\overline{3}_6 \cdot \overline{b}_3\right)\right) \\
&= \overline{s}_6 \cdot \left(\left(\overline{0}_2 \overline{a}_2\right) \otimes \overline{b}_3\right) + \overline{t}_6 \cdot \left(\overline{a}_2 \otimes \left(\overline{0}_3 \overline{b}_3\right)\right) \\
&= \overline{s}_6 \cdot \left(\overline{0}_2 \otimes \overline{b}_3\right) + \overline{t}_6 \cdot \left(\overline{a}_2 \otimes \overline{0}_3\right) \\
&= 0
\end{aligned} \tag{1.2}$$

Hence, this proves that $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z} = 0$. Yet, each $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z} \neq 0$ as R-module, which is a desired counterexample for the problem.

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Problem 2

Let $0 \to M' \xrightarrow{u} M \xrightarrow{v} M'' \to 0$ be an exact sequence of R-modules. Assume that M' and M'' are finitely generated R-modules. Prove that M is a finitely generated R-modules. Explain if the converse is true.

Solution: We'll first prove the given statement, then prove the converse is false in general, by providing a counterexample.

I. Proof of Statement:

Given the sequence is exact, then it implies $M' \xrightarrow{u} M$ is injective, hence $M' \cong \mathrm{im}(u) \subset M$, WLOG we'll denote $\mathrm{im}(u)$ as M', and $M' \xrightarrow{u} M$ as an inclusion map, and recognize all elements in M' as in M also.

Since M', M'' are finitely-generated, there exists $x_1, ..., x_n \in M'$ and $y_1, ..., y_m \in M''$, such that $M' = \sum_{i=1}^{n} Rx_i$ and $M'' = \sum_{j=1}^{m} Ry_j$. Then, by the exactness we have $M \xrightarrow{v} M''$ being surjective, hence we can choose $z_1, ..., z_m \in M$, such that for each index $j \in \{1, ..., m\}$, one has $v(z_j) = y_j$. We claim that the collection $\{x_1, ..., x_n, z_1, ..., z_m\}$ generates M.

For any $x \in M$, since $v(x) \in M''$, there exists $b_1, ..., b_m \in R$, such that the following holds:

$$v(x) = \sum_{j=1}^{m} b_j y_j = \sum_{j=1}^{m} b_j v(z_j) = v\left(\sum_{j=1}^{m} b_j z_j\right) \tag{2.1}$$

As a result, one has $v\left(x - \sum_{j=1}^{m} b_j z_j\right) = 0$, or $x - \sum_{j=1}^{m} b_j z_j \in \ker(v)$.

Again, using exactness of the sequence, one gets $\ker(v) = \mathrm{im}(u) = M'$, hence there exists $a_1, ..., a_n \in R$, such that $x - \sum_{j=1}^{m} b_j z_j = \sum_{i=1}^{n} a_i x_i$ by the finitely generated property of M'.

So, we have $x = \sum_{i=1}^{n} a_i x_i + \sum_{j=1}^{m} b_j z_j$, hence all element $x \in M$ can be expressed as some R-linear combination of $\{x_1, ..., x_n, z_1, ..., z_m\}$, showing M is finitely generated using these elements.

II. Counterexample of the Converse:

Let k be a field, consider the ring $R = k[x_1, x_2, ...]$, which R is a polynomial ring over k, with (countably) infinite indeterminates. First, let's note that R consists of polynomials that are finite sums of monomials, and each monomial can involve only finite indeterminates.

Consider the ideal $I := (x_1, x_2, ...)$ (ideal generated by all indeterminates). Then, $I, R, R/I$ are all R-modules, in particular R is a finitely generated R-module (since every $r \in R$ can be written as $r \cdot 1$, so R is generated by 1 as an R-module). Moreover, we naturally have the following exact sequence of R-linear maps:

$$0 \to I \hookrightarrow R \twoheadrightarrow R/I \to 0$$

And it's simply because the image of the inclusion $I \hookrightarrow R$ is I, and the kernel of projection $R \twoheadrightarrow R/I$ is also I.

Now, we have a short exact sequence, such that the middle R-module is finitely generated. Yet, we claim that I is not a finitely generated R-module:

Suppose the contrary that I is a finitely generated R-module, then there exists $f_1, ..., f_n \in I$, such that $I = \sum_{i=1}^{n} Rf_i = (f_1, ..., f_n)$ as ideal. Based on our definition, there are only finitely many indeterminates involved in $f_1, ..., f_n$, say $S = \left\{x_{i_1}, ..., x_{i_k}\right\}$ (and say it's ordered such that $i_1 < ... < i_k$).

Then, consider the element $x_{i_k+1} \in I$: By our assumption, there exists polynomials $g_1, ..., g_n \in R$, such that $x_{i_k+1} = \sum_{i=1}^n g_i f_i$. Which, let $T = \left\{x_{j_1}, ..., x_{j_l}\right\}$ be the (finite) indeterminates involved in the polynomials $g_1, ..., g_n$. Which, if consider the set of indeterminates $S \cup T \cup \left\{x_{i_k+1}\right\} = \left\{x_{i_k+1}, x_{m_1}, ..., x_{m_p}\right\}$ after reordering, one can realize the elements $x_{i_k+1}, f_1, ..., f_n, g_1, ..., g_n$ as elements in the subring $k\left[x_{i_k+1}, x_{m_1}, ..., x_{m_p}\right] \hookrightarrow k[x_1, x_2, ...]$, and inside this smaller subring, the equality $x_{i_k+1} = \sum_{i=1}^n g_i f_i$ still holds.

Now, consider the evaluation map $k\left[x_{i_k+1}, x_{m_1}, ..., x_{m_p}\right] \to k$, by $x_{i_k+1} \mapsto 1$, and all $x_{m_r} \mapsto 0$: Since we've chosen x_{i_k+1}, such that it's distinct from all x_{i_j} $(1 \leq j \leq k)$, then one has x_{i_j} being some of the x_{m_r}, hence $x_{i_j} \mapsto 0$. Also, notice that since each $f_i \in I$, then it has no constant term (since one has $f_i = h_1 x_1 + ...$ for $h_i \in R$, such that finitely many $h_i \neq 0$; then, all the constant terms in h_i are multiplied with x_i, which is with constant term 0). Hence, with each $x_{i_j} \mapsto 0$, $f_i \mapsto 0$ also (since it's a polynomial involving only $x_{i_1}, ..., x_{i_k}$, and with no constant term).

As a result, one has the following:

$$x_{i_k+1} = \sum_{i=1}^n g_i f_i \mapsto 0 \tag{2.2}$$

because each $f_i \mapsto 0$. However, this contradicts our assumption that $x_{i_k+1} \mapsto 1$. Hence, our assumption must be false, the ideal I cannot be a finitely generated R-module.

Thus, we found an exact sequence of R-modules $0 \to I \hookrightarrow R \twoheadrightarrow R/I \to 0$, such that the middle is finitely generated R-module, while the two sides are not guaranteed to be finitely generated (in particular, the left side is not finitely generated). This proves that the converse is in general false.

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Problem 3

Let M and N be flat R-modules, where R is a commutative ring. Prove or disprove that $M \otimes_R N$ is flat R-module.

Solution: We'll prove that $M \otimes_R N$ is also a flat R-module.

Since the tensor functor $(\_) \otimes_R (M \otimes_R N) : \mathsf{R\text{-}Mod} \to \mathsf{R\text{-}Mod}$ is a right exact functor, to show that $M \otimes_R N$ is flat (or the above tensor functor is exact), one only needs to prove it preserves injective R-linear maps.

Recall that for any R-module K, L, P, there is a unique R-linear isomorphism $\varphi_{KLP} : (K \otimes_R L) \otimes_R P \xrightarrow{\sim} K \otimes_R (L \otimes_R P)$ that satisfies $\varphi_{KLP}((k \otimes l) \otimes p) = k \otimes (l \otimes p)$ for all $k \in K$, $l \in L$, and $p \in P$.

Now, given an injective R-linear map $f : K \hookrightarrow K'$, since M is flat, then the exactness of the functor $(\_) \otimes_R M$ guarantees the map $f \otimes \mathrm{id}_M : K \otimes_R M \to K' \otimes_R M$ is injective (which the map is given by $(f \otimes \mathrm{id}_M)(k \otimes m) = f(k) \otimes m$ for all $k \in K$, $m \in M$).

Similarly, since N is flat, then the exactness of the functor $(\_) \otimes_R N$ guarantees the map $(f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N : (K \otimes_R M) \otimes_R N \to (K' \otimes_R M) \otimes_R N$ is injective (which the map is given by $((f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N)((k \otimes m) \otimes n) = (f(k) \otimes m) \otimes n$ for all $k \in K$, $m \in M$, and $n \in N$).

Finally, consider the following diagram:

$$
\begin{tikzcd}
K \otimes_R (M \otimes_R N) \arrow[r, "f \otimes \mathrm{id}_{M \otimes_R N}"] \arrow[d, hook, two heads, "\varphi^{-1}_{KMN}"'] & K' \otimes_R (M \otimes_R N) \\
(K \otimes_R M) \otimes_R N \arrow[r, hook, "(f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N"'] & (K' \otimes_R M) \otimes_R N \arrow[u, hook, two heads, "\varphi_{K'MN}"']
\end{tikzcd}
$$

Notice that the diagram commutes, because for each $k \in K$, $m \in M$, and $n \in N$, one has the following:

$$\left(f \otimes \mathrm{id}_{M \otimes_R N}\right)(k \otimes (m \otimes n)) = f(k) \otimes (m \otimes n) \tag{3.1}$$

$$
\begin{aligned}
&\varphi_{K'MN} \circ ((f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N) \circ \varphi^{-1}_{KMN}(k \otimes (m \otimes n)) \\
&= \varphi_{K'MN} \circ ((f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N)((k \otimes m) \otimes n) \\
&= \varphi_{K'MN}((f(k) \otimes m) \otimes n) \\
&= f(k) \otimes (m \otimes n)
\end{aligned} \tag{3.2}
$$

Then, since $K \otimes_R (M \otimes_R N)$ is generated by all elements of the form $k \otimes (m \otimes n)$, then the above two calculation shows that the two maps are the same, hence the diagram commutes.

As a result, since $\varphi_{K'MN}$, $(f \otimes \mathrm{id}_M) \otimes \mathrm{id}_N$, and φ^{-1}_{KMN} are all injective, then their composition – which results in $f \otimes \mathrm{id}_{M \otimes_R N}$ – is injective. Therefore, the functor $(\_) \otimes_R (M \otimes_R N)$ preserves injective maps, hence is exact. This shows that $M \otimes_R N$ is a flat module.

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Problem 4

Let $\varphi : R \to S$ be a ring homomorphism and let M be a flat R-module. Prove or disprove that $M \otimes_R S$ is a flat S-module.

Solution: We'll prove that given M is a flat R-module, then $M \otimes_R S$ is a flat S-module.

Since $(M \otimes_R S) \otimes_S (\_) : \mathsf{S\text{-}Mod} \to \mathsf{S\text{-}Mod}$ is right exact, one just needs to prove it preserves injective maps.

For this, let's prove an exercise in Atiyah Macdonald Chapter 2:

Exercise: 2.15

Let A, B be commutative rings, let M be an A-module, P a B-module and N an (A, B)-bimodule (that is, N is simultaneously an A-module and B-module and the two structures are compatible in the sens that $a(xb) = (ax)b$ for all $a \in A$, $b \in B$, $x \in N$).

Then, $M \otimes_A N$ is naturally a B-module, $N \otimes_B P$ an A-module, and we have:

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P) \tag{4.1}$$

Proof: We'll prove the three claims in sequence:

I. B-Module Structure on $M \otimes_A N$:

For any $b \in B$, $m \in M$ and $n \in N$, detine the action $B \times (M \otimes_A N) \to M \otimes_A N$ as $b \cdot (m \otimes_A n) := m \otimes_A (nb)$ on each generator $m \otimes_A n$ (where nb is given by the B-module structure of N), we also define its action on any sum $\sum_{i=1}^k m_i \otimes_A n_i$ by sum of its action on individual $m_i \otimes_A n_i$. As a set map this is well-defined, let's check it does satisfy the B-module properties (which suffices to check for the generators of $M \otimes_A N$). Given any $b, b' \in B$, and any element $m \otimes_A n, m' \otimes_A n' \in M \otimes_A N$, we have the following:

1. $$\begin{aligned}(b + b') \cdot (m \otimes_A n) &= m \otimes_A (n(b + b')) = m \otimes_A (nb + nb') \\ &= m \otimes_A (nb) + m \otimes_A (nb') \\ &= b \cdot (m \otimes_A n) + b' \cdot (m \otimes_A n)\end{aligned} \tag{4.2}$$
2. $$b \cdot (m \otimes_A n + m' \otimes_A n') = b \cdot (m \otimes_A n) + b \cdot (m' \otimes_A n') \tag{4.3}$$

   (Note: This is by our definition, such that the action on the sum, is the sum of each individual generator after the action).
3. $$\begin{aligned}(bb') \cdot (m \otimes_A n) &= m \otimes_A (n(bb')) = m \otimes_A (n(b'b)) \\ &= m \otimes_A ((nb')b) = b \cdot (m \otimes_A (nb')) \\ &= b \cdot (b' \cdot (m \otimes_A n))\end{aligned} \tag{4.4}$$
4. $$1_B \cdot (m \otimes_A n) = m \otimes_A (n1_B) = m \otimes_A n \tag{4.5}$$

These are all the requirements for being a B-module, hence $M \otimes_A N$ has a B-module structure (by its action on N).

As a side note, this in fact makes $M \otimes_A N$ an (A, B)-bimodule structure, since for any $m \otimes_A n \in M \otimes_A N$, $a \in A$, and $b \in B$, we have the following:

$$\begin{aligned}a \cdot (b \cdot (m \otimes_A n)) &= a \cdot (m \otimes_A (nb)) = m \otimes_A (a(nb)) \\ &= m \otimes_A ((an)b) = b \cdot (m \otimes_A (an)) \\ &= b \cdot (a \cdot (m \otimes_A n))\end{aligned} \tag{4.6}$$

Which, the A-action and B-action commutes, showing the bimodule property.

II. A-Module Structure on $N \otimes_B P$:

Similarly, define $A \times (N \otimes_B P) \to N \otimes_B P$ by $a \cdot (n \otimes_B p) := (an) \otimes_B p$. Based on similar reasonings in **I**, it satisfies all the conditions for A-module, hence $N \otimes_B P$ now can be realized as A-module.

Again, similar to the proof in **I**, the computation shows that $N \otimes_B P$ is an (A, B)-bimodule.

III. Isomorphism:

We aim to show that $(M \otimes_A N) \otimes_B P$ and $M \otimes_A (N \otimes_B P)$ are isomorphic as both A and B-module. For this, we'll construct a B-linear map $(M \otimes_A N) \otimes_B P \to M \otimes_A (N \otimes_B P)$, and another A-linear map $(M \otimes_A N) \otimes_B P \leftarrow M \otimes_A (N \otimes_B P)$, and prove that these two maps are isomorphism for abelian groups, while both maps are A-linear and B-linear, which will complete the isomorphism.

1. **B-linear map:**

   First, fix arbitrary $p \in P$, and define a map $F_p : M \times N \to M \otimes_A (N \otimes_B P)$ by $(m, n) \mapsto m \otimes_A (n \otimes_B p)$ for all $(m, n) \in M \times N$. Note that this is A-bilinear, since for all $a, a' \in A$, $m, m' \in M$, and $n, n' \in N$, we have the following:

$$\begin{aligned} F_p(am + a'm', n) &= (am + a'm') \otimes_A (n \otimes_B p) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) + a' \cdot (m' \otimes_A (n \otimes_B p)) \\ &= a \cdot F_p(m, n) + a' \cdot F_p(m, n) \end{aligned} \tag{4.7}$$

$$\begin{aligned} F_p(m, an + a'n') &= m \otimes_A ((an + a'n') \otimes_B p) \\ &= m \otimes_A (a \cdot (n \otimes_B p) + a' \cdot (n' \otimes_B p)) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) + a' \cdot (m \otimes_A (n \otimes_B p)) \\ &= a \cdot F_p(m, n) + a' \cdot F_p(m, n') \end{aligned} \tag{4.8}$$

   Hence, this uniquely factors through the tensor product, as $\overline{F}_p : M \otimes_A N \to M \otimes_A (N \otimes_B P)$ by $\overline{F}_p(m \otimes_A n) = m \otimes_A (n \otimes_B p)$.

   Now, let's define a map $G : (M \otimes_A N) \times P \to M \otimes_A (N \otimes_B P)$ as this:

$$G\left(\sum_{i=1}^{k} m_i \otimes_A n_i \quad , \quad p\right) := \overline{F}_p\left(\sum_{i=1}^{k} m_i \otimes_A n_i\right) \tag{4.9}$$

   We claim that G is B-bilinear (which suffices to check each individual tensors). Given any $b, b' \in B$, $(m \otimes_A n), (m' \otimes_A n') \in M \otimes_A N$, and $p, p' \in P$, we have the following:

$$\begin{aligned}
G(b\cdot(m\otimes_A n)+b'\cdot(m'\otimes_A n'),p) &= G(m\otimes_A (nb)+m'\otimes_A (n'b'),p)\\
&= \overline{F}_p(m\otimes_A (nb)+m'\otimes_A (n'b'))\\
&= m\otimes_A ((nb)\otimes_B p)+m'\otimes_A ((n'b')\otimes_B p)\\
&= m\otimes_A ((n\otimes_B p)b)+m'\otimes_A ((n'\otimes_B p)b')\\
&= b\cdot(m\otimes_A (n\otimes_B p))+b'\cdot(m'\otimes_A (n'\otimes_B p))\\
&= b\cdot\overline{F}_p(m\otimes_A n)+b'\cdot\overline{F}_p(m'\otimes_A n')\\
&= b\cdot G(m\otimes_A n,p)+b'\cdot G(m'\otimes_A n',p)
\end{aligned}\tag{4.10}$$

$$\begin{aligned}
G(m\otimes_A n, bp+b'p') &= F_{bp+b'p'}(m\otimes_A n)\\
&= m\otimes_A (n\otimes_B (bp+b'p'))\\
&= m\otimes_A (b\cdot(n\otimes_B p)+b'\cdot(n\otimes_B p'))\\
&= b\cdot(m\otimes_A (n\otimes_B p))+b'\cdot(m\otimes_A (n\otimes_B p'))\\
&= b\cdot\overline{F}_p(m\otimes_A n)+b'\cdot\overline{F}_{p'}(m\otimes_A n)\\
&= b\cdot G(m\otimes_A n,p)+b'\cdot G(m\otimes_A n,p')
\end{aligned}\tag{4.11}$$

Since G is B-bilinear, there exists a unique B-linear map $\overline{G}:(M\otimes_A N)\otimes_B P\to M\otimes_A (N\otimes_B P)$ by $\overline{G}((m\otimes_A n)\otimes_B p)=m\otimes_A (n\otimes_B p)$.

2. ***A*-linear map:**

   Similarly, fix arbitrary $m\in M$, and define a map $H_m : N\times P\to (M\otimes_A N)\otimes_B P$ by $(n,p)\mapsto (m\otimes_A n)\otimes_B p$ for all $(n,p)\in N\times P$. Using the same logic in the previous part, one can prove that H_m is B-bilinear, hence inducing a unique B-linear map $\overline{H}_m : N\otimes_B P\to (M\otimes_A N)\otimes_B P$ by $\overline{H}_m(n\otimes_B p)=m\otimes_A (n\otimes_B p)$.

   Then, one defines another map $L: M\times(N\otimes_B P)\to (M\otimes_A N)\otimes_B P$ as follow:

$$L\left(m,\sum_{i=1}^k n_i\otimes_B p_i\right):=\overline{H}_m\left(\sum_{i=1}^k n_i\otimes_B p_i\right)\tag{4.12}$$

   Similar to the previous part, we in fact have L being A-bilinear after running through the computation. Hence, this induces a unique A-linear map $\overline{L}: M\otimes_A (N\otimes_B P)\to (M\otimes_A N)\otimes_B P$ by $\overline{L}(m\otimes_A (n\otimes_B p))=(m\otimes_A n)\otimes_B p$.

3. **Isomorphism:**

   Given the B-linear map $\overline{G}:(M\otimes_A N)\otimes_B P\to M\otimes_A (N\otimes_B P)$, and the A-linear map $\overline{L}: M\otimes_A (N\otimes_B P)\to (M\otimes_A N)\otimes_B P$, if we first look at the two modules' abelian group structure, one have this for all $m\in M, n\in N, p\in P$:

$$\begin{aligned}
\overline{L}\circ\overline{G}((m\otimes_A n)\otimes_B p) &= \overline{L}(m\otimes_A (n\otimes_B p))=(m\otimes_A n)\otimes_B p\\
\overline{G}\circ\overline{L}(m\otimes_A (n\otimes_B p)) &= \overline{G}((m\otimes_A n)\otimes_B p)=m\otimes_A (n\otimes_B p)
\end{aligned}\tag{4.13}$$

   Hence, as homomorphism of abelian grouops, $\overline{G},\overline{L}$ are mutual inverses (since the compositions act as identity on all the generators of the two spaces, in the right composition order).

   Now, it suffices to check that $\overline{G},\overline{L}$ are in fact both A-linear and B-linear.

   - For $\overline{G}$, we've constructed it as a B-linear map, so it suffices to prove the A-linear structure (in particular, suffices to prove it preserves the A-action, since additivity is given by the fact that $\overline{G}$ is a homomorphism of abelian group). Given any $a \in A$, and $(m \otimes_A n) \otimes_B p \in (M \otimes_A N) \otimes_B P$, one has the following:

$$\begin{aligned} \overline{G}(a \cdot ((m \otimes_A n) \otimes_B p)) &= \overline{G}((a \cdot (m \otimes_A n)) \otimes_B p) \\ &= \overline{G}(((am) \otimes_A n) \otimes_B p) \\ &= (am) \otimes_A (n \otimes_B p) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) \\ &= a \cdot \overline{G}((m \otimes_A n) \otimes_B p) \end{aligned} \tag{4.14}$$

(Note: Here we use the fact that $(M \otimes_A N) \otimes_B P$ has an A-module structure by letting A act on $M \otimes_A N$, which is proven in **I**, because M, N can be arbitrary).

Hence, $\overline{G}$ preserves A-action, which is A-linear.

- Similarly, for $\overline{L}$, it's constructed as an A-linear map, so it sufices to prove it preserves the B-action. Given any $b \in B$, and $m \otimes_A (n \otimes_B p) \in M \otimes_A (N \otimes_B P)$, one has the following:

$$\begin{aligned} \overline{L}(b \cdot (m \otimes_A (n \otimes_B p))) &= \overline{L}(m \otimes_A (b \cdot (n \otimes_B p))) \\ &= \overline{L}(m \otimes_A (n \otimes_B (bp))) \\ &= (m \otimes_A n) \otimes_B (bp) \\ &= b \cdot ((m \otimes_A n) \otimes_B p) \\ &= b \cdot \overline{L}(m \otimes_A (n \otimes_B p)) \end{aligned} \tag{4.15}$$

(Note: Again, we use the fact that $M \otimes_A (N \otimes_B P)$ has B-module structure by letting B act on $N \otimes_B P$, proven in **II**).

Hence, $\overline{L}$ preserves B-action, which is a B-linear map.

This realizes $\overline{G}, \overline{L}$ as mutual inverses as both A-linear and B-linear map, hence proving that $(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P)$, as (A, B)-bimodule. In particular, an isomorphism (as both A and B-module) between them that's constructed above, has the following form:

$$\begin{aligned} &\varphi_{MNP} : (M \otimes_A N) \otimes_B P \xrightarrow{\sim} M \otimes_A (N \otimes_B P) \\ &\varphi_{MNP}((m \otimes_A n) \otimes_B p) = m \otimes_A (n \otimes_B p), \quad \forall m \in M, n \in N, p \in P \end{aligned} \tag{4.16}$$

$\square$

Since S is an (R, S)-bimodule (where it's both an R-module and S-module, while any $r \in R$, $s, t \in S$ has $(r \cdot t)s = (\varphi(r)t)s = \varphi(r)(ts) = r \cdot (ts)$, where the dot $\cdot$ represents the R-action on S), then one has the following isomorphism, for any S-module P:

$$\varphi_{MSP} : (M \otimes_R S) \otimes_S P \xrightarrow{\sim} M \otimes_R (S \otimes_S P), \quad \varphi_{MSP}((m \otimes_R s) \otimes_S p) = m \otimes_R (s \otimes_S p) \tag{4.17}$$

Now, given any injective S-linear map $f : P \hookrightarrow P'$ between two S-modules, one simply has $(\mathrm{id}_S) \otimes_S f : S \otimes_B P \hookrightarrow S \otimes_B P'$ be injective (since tensor an S-module over S is isomorphic to the original module).

However, $(\mathrm{id}_S) \otimes_S f$ is also an injective R-linear map, since it's additive, and for all $r \in R$, $s \in S$, and $p \in P$, one has the following:

$$
\begin{aligned}
((\mathrm{id}_S) \otimes_S f)(r \cdot (s \otimes_B p)) &= ((\mathrm{id}_S) \otimes_S f)((r \cdot s) \otimes_B p) \\
&= (r \cdot s) \otimes_B f(p) \\
&= r \cdot (s \otimes_B f(p)) \\
&= r \cdot ((\mathrm{id}_S) \otimes_S f)(s \otimes_B p)
\end{aligned} \tag{4.18}
$$

Hence, with M being a flat R-module, the R-linear map $\mathrm{id}_M \otimes_R ((\mathrm{id}_S) \otimes_S f) : M \otimes_R (S \otimes_S P) \hookrightarrow M \otimes_R (S \otimes_S P')$ is also injective.

Similar to the previous one, we also have $\mathrm{id}_M \otimes_R ((\mathrm{id}_S) \otimes_S f)$ being an injective S-linear map, since it's additive, and also for any $m \in M$, $s, t \in S$, and $p \in P$, one has the following:

$$
\begin{aligned}
(\mathrm{id}_M \otimes_R ((\mathrm{id}_S) \otimes_S f))(s \cdot (m \otimes_R (t \otimes_S p))) &= (\mathrm{id}_M \otimes_R ((\mathrm{id}_S) \otimes_S f))(m \otimes_R (s \cdot (t \otimes_S p))) \\
&= m \otimes_R (((\mathrm{id}_S) \otimes_S f)((st) \otimes_S p)) \\
&= m \otimes_R ((st) \otimes_S f(p)) \\
&= m \otimes_R (s \cdot (t \otimes_S f(p))) \\
&= s \cdot (m \otimes_R (t \otimes_S f(p))) \\
&= s \cdot [(\mathrm{id}_M \otimes_R ((\mathrm{id}_S) \otimes_S f))(m \otimes_R (t \otimes_S p))]
\end{aligned} \tag{4.19}
$$

For simplicity, let's define $\psi := \mathrm{id}_M \otimes_R ((\mathrm{id}_S) \otimes_S f)$ as the injective S-linear map.

Finally, one has the following diagram:

$$
\begin{tikzcd}
(M \otimes_R S) \otimes_S P \arrow[r, "(\mathrm{id}_M \otimes_R S) \otimes_S f"] \arrow[d, hook, two heads, "\varphi_{MSP}"'] & (M \otimes_R S) \otimes_S P' \\
M \otimes_R (S \otimes_S P) \arrow[r, hook, "\psi"'] & M \otimes_R (S \otimes_S P') \arrow[u, hook, two heads, "\varphi^{-1}_{MSP'}"']
\end{tikzcd}
$$

Notice that this big diagram commutes, because for all $m \in M$, $s \in S$, and $p \in P$, one has the following:

$$
((\mathrm{id}_M \otimes_R S) \otimes_S f)((m \otimes_R s) \otimes_S p) = (m \otimes_R s) \otimes_S f(p) \tag{4.20}
$$

$$
\begin{aligned}
\varphi^{-1}_{MSP'} \circ \psi \circ \varphi_{MSP}((m \otimes_R s) \otimes_S p) &= \varphi^{-1}_{MSP'} \circ \psi(m \otimes_R (s \otimes_S p)) \\
&= \varphi^{-1}_{MSP'}(m \otimes_R (s \otimes_S f(p))) \\
&= (m \otimes_R s) \otimes_S f(p)
\end{aligned} \tag{4.21}
$$

This shows that $\varphi^{-1}_{MSP'} \circ \psi \circ \varphi_{MSP} = (\mathrm{id}_M \otimes_R S) \otimes_S f$ as S-linear maps. Hence, with each map on the LRS being injective, the RHS $(\mathrm{id}_M \otimes_R S) \otimes_S f$ is also injective.

This shows that $(M \otimes_R S) \otimes_S (\_)$ is an exact functor (as it preserves injectivity), hence $M \otimes_R S$ can be realized as a flat S-module.

5 D

Problem 5

Let $\varphi : R \to S$ be a ring homomorphism such that M is projective as an R-module. Prove or disprove that $M \otimes_R S$ is projective as S-module.

Solution: We'll prove that if M is projective as R-module, then $M \otimes_R S$ is projective as S-module.

Since M is projective as an R-module, there exists another R-module N, such that $M \oplus N \cong \bigoplus_{i\in I} Re_i$ a free R-module (where each $Re_i \cong R$ as R-module). Then, since tensor product „commutes“ with direct sum (up to isomorphism), hence one has the following isomorphism as R -modules:

$$(M \otimes_R S) \oplus (N \otimes_R S) \cong (M \oplus N) \otimes_R S \cong \left(\bigoplus_{i\in I} Re_i\right) \otimes_R S \cong \bigoplus_{i\in I} (Re_i \otimes_R S) \tag{5.1}$$

As a remark, all these isomorphism respects the S-module structure: Given any family of R-modules $\{M_i\}_{i\in\Lambda}$, the R-linear isomorphism $\varphi : \left(\bigoplus_{i\in\Lambda} M_i\right) \otimes_R S \xrightarrow{\sim} \bigoplus_{i\in I} (M_i \otimes_R S)$ is given as:

$$\varphi\Big(\big((m_i)_{i\in\Lambda}\big) \otimes s\Big) = (m_i \otimes s)_{i\in\Lambda} \tag{5.2}$$

Which, this map is also S-linear, as any $(m_i)_{i\in\Lambda} \in \bigoplus_{i\in\Lambda} M_i$, and $s, t \in S$, satisfy the following:

$$\begin{aligned} \varphi\Big(s \cdot \big((m_i)_{i\in\Lambda} \otimes t\big)\Big) &= \varphi\Big((m_i)_{i\in\Lambda} \otimes (ts)\Big) = (m_i \otimes (ts))_{i\in\Lambda} \\ &= (s \cdot (m_i \otimes t))_{i\in\Lambda} = s \cdot (m_i \otimes t)_{i\in\Lambda} \end{aligned} \tag{5.3}$$

Hence, φ is also an S-linear isomorphism. Apply this logic to the 1st and 3rd isomorphism relation, $(M \otimes_R S) \oplus (N \otimes_R S) \cong \bigoplus_{i\in I} (Re_i \otimes_R S)$ as S-module.

Now, look at each $Re_i \cong R$, one has each $Re_i \otimes_R S \cong S$ as R-module (and a specific R-linear isomorphism is given by $\varphi : R \otimes_R S \xrightarrow{\sim} S$, $\varphi(r \otimes s) = r \cdot s = \varphi(r)s$).

However, notice that this φ is also an S-linear map, because it's additive, and also every $r \in R$ and $s, t \in S$ satisfy the following:

$$\varphi(s \cdot (r \otimes t)) = \varphi(r \otimes (ts)) = r \cdot (ts) = \varphi(r)(ts) = (\varphi(r)t)s = s \cdot (r \cdot t) = s \cdot \varphi(r \otimes t) \tag{5.4}$$

(Note: Here the S-module structure on $R \otimes_R S$ is by letting S act on S).

Hence, it's not only an R-linear isomorphism, but in fact an S-linear isomorphism. As a consequence, with the S-module structure on $R \otimes_R S$ being isomorphic to S, one has:

$$(M \otimes_R S) \oplus (N \otimes_R S) \cong \bigoplus_{i\in I} (Re_i \otimes_R S) \cong \bigoplus_{i\in I} S \tag{5.5}$$

Where the final part is a free S-module. This realizes $M \otimes_R S$ as a direct summand of a free S-module, hence it's projective as S-module.

6 D

Problem 6

Let $\varphi : R \to S$ be a ring homomorphism and let M be an injective R-module such that $M \otimes_R S \neq 0$. Prove or disprove that $M \otimes_R S$ is injective S-module.

Solution: We'll disprove the statement, by providing a counterexample.

Consider the ring $R = \mathbb{Z}$, $S = \mathbb{Z}[x]$, and the ring homomorphism $\varphi : \mathbb{Z} \hookrightarrow \mathbb{Z}[x]$ as the inclusion. Let $M := \mathbb{Q}$ be the injective $\mathbb{Z}$-module (by the divisibility, and the equivalence of divisible and injective modules over $\mathbb{Z}$, a PID). Here, we claim that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}[x]$ is not an injective $\mathbb{Z}[x]$-module.

Consider the fact that tensor product of two commutative R-algebra serves as fibre coproduct in the category of commutative rings, in this category we have the following diagram:

$$\begin{tikzcd}
\mathbb{Z} \arrow[r, hook, "\iota_{\mathbb{Q}}"] \arrow[d, hook, "\iota_x"'] & \mathbb{Q} \arrow[d, hook, "\alpha"] \\
\mathbb{Z}[x] \arrow[r, hook, "\beta"'] & \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}[x]
\end{tikzcd}$$

Where, any $q \in \mathbb{Q}$ satisfies $\alpha(q) = q \otimes 1$, and any integer polynomial $f(x) \in \mathbb{Z}[x]$ satisfies $\beta(f(x)) = 1 \otimes f(x)$.

Then, consider the ring $\mathbb{Q}[x]$, we claim that as $\mathbb{Z}$-algebra it's isomorphic to $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}[x]$: It has two inclusions $\iota_1 : \mathbb{Q} \hookrightarrow \mathbb{Q}[x]$, and $\iota_2 : \mathbb{Z}[x] \hookrightarrow \mathbb{Q}[x]$, and both inclusions preserve $\mathbb{Z}$ element wise (hence $\iota_1 \circ \iota_{\mathbb{Q}} = \iota_2 \circ \iota_x$). Then, using the universality of fibre coproduct, one realizes a unique $\mathbb{Z}$-algebra homomorphism $h : \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}[x] \to \mathbb{Q}[x]$, such that the following holds:

$$\begin{tikzcd}
\mathbb{Z} \arrow[r, hook, "\iota_{\mathbb{Q}}"] \arrow[d, hook, "\iota_x"'] & \mathbb{Q} \arrow[d, hook, "\alpha"] \arrow[ddr, hook, bend left, "\iota_1"] & \\
\mathbb{Z}[x] \arrow[r, hook, "\beta"] \arrow[drr, hook, bend right, "\iota_2"'] & \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}[x] \arrow[dr, dotted, "\exists! h"] & \\
 & & \mathbb{Q}[x]
\end{tikzcd}$$

In particular, the description is given by $h(q \otimes f(x)) = \iota_1(q)\iota_2(f(x)) = qf(x)$.

Now, let's construct an inverse: Define a map $j : \mathbb{Q}[x] - \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}[x]$, such that every monomial $qx^n \in \mathbb{Q}[x]$ has $j(qx^n) := q \otimes x^n$, and for general polynomials $f(x) = \sum_{i=0}^n q_i x^i$, it satisfies $j(f(x)) := \sum_{i=0}^n q_i \otimes x^i$. To verify this is a well-defined $\mathbb{Z}$-algebra homomorphism, given any $f(x) = \sum_{i=0}^n q_i x^i$ and $g(x) = \sum_{j=0}^m p_j x^j$ in $\mathbb{Q}[x]$ (WLOG, say $n \geq m$), one has the following:

$$\begin{aligned}
j(f(x)+g(x)) &= j\left(\sum_{j=0}^{m}(q_j+p_j)x^j + \sum_{i=m+1}^{n} q_i x^i\right) \\
&= \sum_{j=0}^{m}(q_j+p_j)\otimes x^j + \sum_{i=m+1}^{n} q_i \otimes x^i \\
&= \sum_{i=0}^{m} q_i\otimes x^i + \sum_{i=m+1}^{n} q_i\otimes x^i + \sum_{j=0}^{m} p_j\otimes x^j \\
&= \sum_{i=0}^{n} q_i\otimes x^i + \sum_{j=0}^{m} p_j\otimes x^j \\
&= j(f(x)) + j(g(x))
\end{aligned} \tag{6.1}$$

This proves the additivity of j. Also, we have the following:

$$\begin{aligned}
j(f(x)g(x)) &= j\left(\sum_{i=0}^{n}\sum_{j=0}^{m} q_i p_j x^{i+j}\right) \\
&= \sum_{i=0}^{n}\sum_{j=0}^{m}(q_i p_j)\otimes x^{i+j} \\
&= \sum_{i=0}^{n}\sum_{j=0}^{m}(q_i\otimes x^i)\cdot(p_j\otimes x^j) \\
&= \left(\sum_{i=0}^{n} q_i\otimes x^i\right)\left(\sum_{j=0}^{m} p_j\otimes x^j\right) \\
&= j(f(x))\cdot j(g(x))
\end{aligned} \tag{6.2}$$

This proves j is multiplicative, hence it's a well-defined $\mathbb{Z}$-algebra homomorphism.

Now, notice that h, j are mutual inverses of each other: Given any $q\otimes x^n \in \mathbb{Q}\otimes_{\mathbb{Z}}\mathbb{Z}[x]$, and $qx^n \in \mathbb{Q}[x]$ (the generators of each algebra), one has the following:

$$j\circ h(q\otimes x^n) = j(qx^n) = q\otimes x^n, \quad h\circ j(qx^n) = h(q\otimes x^n) = qx^n \tag{6.3}$$

Since both compositions act as identity on the generators of each algebra, they're mutual inverses. Hence, it proves $\mathbb{Q}\otimes_{\mathbb{Z}}\mathbb{Z}[x]\cong\mathbb{Q}[x]$ as $\mathbb{Z}$-algebra. Moreover, it's $\mathbb{Z}[x]$-module structure is given by the inclusion $\iota_2 : \mathbb{Z}[x]\hookrightarrow\mathbb{Q}[x]$.

Finally, realize that $\mathbb{Q}[x]$ is not an injective $\mathbb{Z}[x]$-module: Given the element $x\in\mathbb{Z}[x]$, the multiplication map $x\cdot(\_) : \mathbb{Q}[x]\to\mathbb{Q}[x]$ has all $f(x)\in\mathbb{Q}[x]$ satisfies $f(x)\mapsto xf(x)$. In particular, if $f(x)\neq 0$, one has $\deg(xf(x))\geq\deg(x)\geq 1$. Hence, the image of $x\cdot(\_)$ doesn't contain any nonzero constant polynomial, showing it's not surjective.

Hence, as a $\mathbb{Z}[x]$-module, $\mathbb{Q}[x]\cong\mathbb{Q}\otimes_{\mathbb{Z}}\mathbb{Z}[x]$ is not divisible, hence not injective.

With $R=\mathbb{Z}$, $S=\mathbb{Z}[x]$ (with module structure induced by inclusion $\varphi:\mathbb{Z}\hookrightarrow\mathbb{Z}[x]$), and $M=\mathbb{Q}$ an injective $\mathbb{Z}$-module, we have $M\otimes_R S\cong\mathbb{Q}[x]$ as R-algebra. Yet, it is not an injective S-module, which provides a counterexample to the statement.

7 D

Problem 7

Let $\varphi : R \to S$ be a ring homomorphism such that S is flat R-algebra. Prove or disprove that φ is injective.

Solution: We'll disprove it by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, recall that it has a projection $\varphi : R \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ by $\varphi(\overline{1}_6) = \overline{1}_3$, this realizes $\mathbb{Z}/3\mathbb{Z}$ as an R-algebra, while the map φ is not injective (since $\varphi(\overline{3}_6) = \overline{3}_3 = 0$).

Now, consider the ideal $I = \{\overline{0}_6, \overline{2}_6, \overline{4}_6\} \subset R$, which it can be obtained using another projection $\pi_2 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ by $\pi_2(\overline{n}_6) = \overline{n}_2$. Which, $\overline{n}_2 = \overline{0}_2$ iff n is odd iff $\overline{n}_6 \in \{\overline{0}_6, \overline{2}_6, \overline{4}_6\}$, showing $\ker(\varphi) = I$.

However, this ideal also has a ring structure: It's closed under addition and multiplication by the ideal property, so it suffices to find the multiplicative identity. Consider the following multiplication relation:

$$\overline{0}_6 \cdot \overline{4}_6 = \overline{0}_6, \quad \overline{2}_6 \cdot \overline{4}_6 = \overline{8}_6 = \overline{2}_6, \quad \overline{4}_6 \cdot \overline{4}_6 = \overline{16}_6 = \overline{4}_6 \tag{7.1}$$

This shows that all $\overline{a}_6 \in I$ satisfies $\overline{a}_6 \cdot \overline{4}_6 = \overline{a}_6$, which $\overline{4}_6$ is the multiplicative identity of I.

Notice that I as a ring with the above structure is isomorphic to $\mathbb{Z}/3\mathbb{Z}$: As an additive abelian group because both I and $\mathbb{Z}/3\mathbb{Z}$ has order 3, they must be isomorphic, and one abelian group isomorphism can be considered as $j : \mathbb{Z}/3\mathbb{Z} \xrightarrow{\sim} I$, $j(\overline{1}_3) = \overline{4}_6$ (since both are cyclic with prime order, then any nonzero element is a generator), implying $j(\overline{2}_3) = \overline{2}_6$.

Yet, j is also a ring homomorphism, because of the following (for anything multiplied by zero, it's sent to zero by group homomorphism property, so we don't need to check):

$$j(\overline{1}_3 \cdot \overline{2}_3) = j(\overline{2}_3) = \overline{2}_6 = \overline{4}_6 \cdot \overline{2}_6 = j(\overline{1}_3) \cdot j(\overline{2}_3) \tag{7.2}$$

$$j(\overline{1}_3 \cdot \overline{1}_3) = j(\overline{1}_3) = \overline{4}_6 = \overline{4}_6 \cdot \overline{4}_6 = j(\overline{1}_3) \cdot j(\overline{1}_3) \tag{7.3}$$

$$j(\overline{2}_3 \cdot \overline{2}_3) = j(\overline{1}_3) = \overline{4}_6 = \overline{2}_6 \cdot \overline{2}_6 = j(\overline{2}_3) \cdot j(\overline{2}_3) \tag{7.4}$$

Hence, this realizes $\mathbb{Z}/3\mathbb{Z} \cong I$ as rings, and even stronger – as R-algebras. This suffices to check the R-action on $\overline{1}_3$ (as the rest follows directly by adding $\overline{1}_3$ to itself):

$$\begin{aligned}
j(\overline{1}_6 \cdot \overline{1}_3) = j(\overline{1}_3) = \overline{4}_6 = \overline{1}_6 \cdot \overline{4}_6 = \overline{1}_6 \cdot j(\overline{1}_3)\\
j(\overline{2}_6 \cdot \overline{1}_3) = j(\overline{2}_3) = \overline{2}_6 = \overline{2}_6 \cdot \overline{4}_6 = \overline{2}_6 \cdot j(\overline{1}_3)\\
j(\overline{3}_6 \cdot \overline{1}_3) = j(\overline{0}_3) = \overline{0}_6 = \overline{3}_6 \cdot \overline{4}_6 = \overline{3}_6 \cdot j(\overline{1}_3)\\
j(\overline{4}_6 \cdot \overline{1}_3) = j(\overline{1}_3) = \overline{4}_6 = \overline{4}_6 \cdot \overline{4}_6 = \overline{4}_6 \cdot j(\overline{1}_3)\\
j(\overline{5}_6 \cdot \overline{1}_3) = j(\overline{2}_3) = \overline{2}_6 = \overline{5}_6 \cdot \overline{4}_6 = \overline{1}_6 \cdot j(\overline{1}_3)
\end{aligned} \tag{7.5}$$

This proves the R-linearity of j, hence $j : \mathbb{Z}/3\mathbb{Z} \xrightarrow{\sim} I$ is in fact an R-linear isomorphism.

As a result, this is actually a section of the map $\varphi : R \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ when viewed as an R-linear map:

$$\varphi \circ j(\overline{1}_3) = \varphi(\overline{4}_6) = \overline{1}_3 \tag{7.6}$$

And, with $\overline{1}_3$ being a generator of $\mathbb{Z}/3\mathbb{Z}$ additive wise, this proves that $\varphi \circ j = \mathrm{id}_{\mathbb{Z}/3\mathbb{Z}}$. Hence, the R-linear map $\varphi : R \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ splits, showing that $\mathbb{Z}/3\mathbb{Z}$ is a direct summand of R (which is a free R-module).

This proves that $\mathbb{Z}/3\mathbb{Z}$ is a projective R-module, which in particular is flat. However, as a flat R-algebra, the map $\varphi : R \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ is not injective, which provides a counterexample to our statement.

8 D

Problem 8

Let $\varphi : R \to S$ be an injective ring homomorphism such that S is a field. Prove or disprove that S is a flat R-algebra.

Solution: We'll prove that S is a flat R-algebra.

First, since $\varphi : R \hookrightarrow S$ is injective, we can identify R as a subring of S, and φ as a simple inclusion. As a side note, R is an integral domain.

Now, consider any inclusion of nonzero ideal $\iota : I \hookrightarrow R$, and consider its map after tensoring with S over R, say $\iota \otimes \mathrm{id}_S : I \otimes_R S \to R \otimes_R S$. The goal is to proof this map is injective (in fact, we'll prove that it's an isomorphism): Consider the isomorphism as R-modules $R \otimes_R S \xrightarrow{\sim} S$ by $r \otimes s \mapsto r$. Then, $\iota \otimes \mathrm{id}_S$ can also be identified as follow:

$$\iota \otimes \mathrm{id}_S : I \otimes_R S \to S, \quad (\iota \otimes \mathrm{id}_S)(a \otimes s) = as \tag{8.1}$$

Which, fix a nonzero $a \in I$, define the map $g : S \to I \otimes_R S$ by $g(s) := a \otimes \frac{s}{a}$. Notice that this is well-defined, as for any nonzero $a, b \in I$ and any $s \in S$, one has the following:

$$a \otimes \frac{s}{a} = a \otimes \frac{bs}{ba} = (ab) \otimes \frac{s}{ba} = b \otimes \frac{as}{ba} = b \otimes \frac{s}{b} \tag{8.2}$$

Hence, the image is in fact independent of the choice in I. Now, notice that g is also an R-linear map. For all $r \in R$ and $s, t \in S$, one has the following:

$$g(s+t) = a \otimes \frac{s+t}{a} = a \otimes \left(\frac{s}{a} + \frac{t}{a}\right) = a \otimes \frac{s}{a} + a \otimes \frac{t}{a} = g(s) + g(t) \tag{8.3}$$

$$g(r \cdot s) = a \otimes \frac{rs}{a} = r \cdot \left(a \otimes \frac{s}{a}\right) = r \cdot g(s) \tag{8.4}$$

Finally, recognize that g is a mutual inverse of $\iota \otimes \mathrm{id}_S$, as for all $s \in S$ and $b \in I$, one has the following;

$$g \circ (\iota \otimes \mathrm{id}_S)(b \otimes s) = g(bs) = a \otimes \frac{bs}{a} = (ab) \otimes \frac{s}{a} = b \otimes \frac{as}{a} = b \otimes s \tag{8.5}$$

(Note: since $b \in I$, then with ab on the left one can factor out a, and still have $b \otimes \frac{as}{a} \in I \otimes_R S$).

$$(\iota \otimes \mathrm{id}_S) \circ g(s) = (\iota \otimes \mathrm{id}_S)\left(a \otimes \frac{s}{a}\right) = a \cdot \frac{s}{a} = s \tag{8.6}$$

This shows that $\iota \otimes \mathrm{id}_S$, g are mutual inverses on the generators of $I \otimes_R S$ and S, hence are mutual inverses on the two R-modules.

This proves injectivity of $\iota \otimes \mathrm{id}_S$. And, since $I \hookrightarrow R$ is arbitrary inclusion of nonzero ideals (while the zero ideal case is trivial), this implies S is a flat R-algebra.

9 D

Problem 9

Let R be a dedekind domain and let $I \subset R$ be an ideal. Prove or disprove that I is a flat R-module.

Solution: We'll prove that I is a flat R-module.

Recall that flatness is a local property, given any R-module M, one has M being R-flat $\iff$ M_P is R_P-flat for all prime ideal $P \subseteq R$.

For this, let's recall some important properties for dedekind domain, that its Noetherian, all proper ideal factors into primes, and every prime ideal is maximal. Using these two properties, one has the following:

Lemma

Given R a dedekind domain, any prime ideal $P \subseteq R$ has the localization R_P being a PID.

Proof: Let's fix a prime ideal $P \subseteq R$, take any ideal $J \subseteq R_P$. Recall that $J^{ce} = J$ (the property of localization in general).

Look at the ideal $J^c \subseteq R$, by the ideal factorization property of dedekind domain, one has $J^c = Q_1 ... Q_n$, where each $Q_i \subseteq R$ is a prime (hence maximal) ideal. Then, since extension of ideals under localization preserves products of ideals, one has the following:

$$J = J^{ce} = (Q_1 ... Q_n)^e = Q_1^e ... Q_n^e \tag{9.1}$$

Now, notice that for any ideal $I \subseteq R$, one has $I^e \neq R_P$ iff $I \cap R \setminus P = \emptyset$, or I^e is not unit ideal iff $I \subseteq P$. Hence, for any prime ideal Q_i mentioned above, one has $Q_i^e \neq R_P$ iff it's a prime ideal contained in P; yet, since all prime ideals in R is maximal, $Q_i^e \neq R_P$ iff $Q_i \subseteq P$ iff $Q_i = P$. So, one has $J = (P^e)^k$ for some $k \in \mathbb{N}$ (where k indicates the number of prime ideals $Q_i = P$).

Then, since $P^e \subseteq R_P$ is the maximal ideal, the above shows all ideals of R_P is a power of its maximal ideal. Hence, it suffices to prove the maximal ideal P^e is principal (as $P^e = (a)$ implies $(P^e)^l = (a^l)$ for any $l \in \mathbb{N}$). There are two cases to consider:

- If $(P^e)^2 = P^e$, then when realizing P^e as an R_P-module (which is finitely generated, since $P \subseteq R$ is finitely generated). Apply Nakayama's Lemma, because R_P is a local ring with maximal ideal P^e, $(P^e)^2 = P^e$ forces $P^e = 0$. Hence, R_P is a field, which is trivially a PID.

- Else if $(P^e)^2 \subsetneq P^e$, take any $r \in P^e \setminus (P^e)^2$, we claim that $(r) = P^e$: By the factorization above, $(r) = (P^e)^n$ for some $n \in \mathbb{N}$, however if $n \geq 2$ one has $(r) = (P^e)^n \subseteq (P^e)^2$, contradicting the assumption $r \notin (P^e)^2$. Also, $n \neq 0$ as $(r) = (P^e)^0 = R$ is another contradiction to $(r) \subseteq P^e$ (since P^e is not unit ideal). So, $n = 1$, and $(r) = P^e$. This shows that P^e is a principal ideal, implying R_P is a PID.

□

Now, notice that for any ideal $I \subseteq R$, one has the localization I_P being the extension I^e under the map $\varphi : R \to R_P$, which $I^e = \left(\frac{a}{s}\right)$ for some $\frac{a}{s} \in R_P$ by the previous lemma. Hence, as R-module $I^e = \left(\frac{a}{s}\right) \cong R_P$ via the map $I^e \to R_P$ by $\frac{a}{s} \mapsto 1$, showing I^e is a free R_P-module, hence flat R_P-module.

This shows the localization I_P around any prime ideal is flat R_P-module, hence I itself must be flat R-module.

10 D

Problem 10

Let R be a PID and let M be a finitely generated flat R-module. Prove or disprove that M is a free R-module.

Solution: We'll prove that M is a free R-module.

Given that M is a finitely generated flat R-module, define $n :=$ minimum number of generators needed for M, and use $x_1, ..., x_n \in M$ to denote a set of generators of M with minimum length. We'll do induction on the number n.

For the base case $n = 1$, if $M = Rx_1$, then there exists a surjective R-linear map $\varphi : R \twoheadrightarrow M$, given by $\varphi(r) := rx_1$. Then, as R-module, one has $R/\ker(\varphi) \cong M = Rx_1$, given by $\overline{r} \mapsto rx_1$. However, based on the flatness of M, this enforces $\ker(\varphi) = 0$: Suppose the contrary $\ker(\varphi) \neq 0$, there exists nonzero element $a \in \ker(\varphi)$, as a consequence $ax_1 = 0$ (since under the map $R/\ker(\varphi) \to M$, one has $\overline{0} = \overline{a} \mapsto ax_1$). Then, the multiplication map $a \cdot (\_) : M \to M$ is not injective, showing M is torsion. Yet, this contradicts the flatness condition of M, hence one must have $\ker(\varphi) = 0$, further implying $M \cong R/\ker(\varphi) \cong R$. So, M is a free R-module with rank 1.

Now, suppose for all case $k < n$, M having minimum number of k generators implies M is a free R-module, then consider the case for n, one has $M = \sum_{i=1}^{n} Rx_i$. Define $N := \sum_{i=1}^{n-1} Rx_i$, by the minimality of number of generators, one must have $x_n \notin N$ (or else $M = N$ and hence only needs $(n-1)$ number of generators, contradicting the minimality of n).

Here, we claim that $Rx_n \cap N = 0$, and hence $M \cong N \oplus Rx_n$:

Suppose the contrary that the intersection is nonzero, there exists some $a \in R$, such that $ax_n \in N$ (and with $x_n \notin N$, a is non-unit). Define the ideal $I := \{a \in R \mid ax_n \in N\}$, then by the PID property, one has $I = (b)$ for some nonzero $b \in R$. On the other hand, I is not the unit ideal.

Then, define the multiplication map $f = b \cdot (\_) : M \to M$. Since any $m \in M$ has $f(m) = bm \in IM$, $f(M) \subseteq IM$, with the assumption M is finitely generated, one can apply „Cayley-Hamilton Theorem" covered in class, there exists some $k \in \mathbb{N}$, and some $c_1, ..., c_k \in R$ (or $c_1 b, ..., c_k b \in I = (b)$), such that the endomorphism $f^k + c_1 b f^{k-1} + ... c_{k-1} b f + c_k b = 0$ on M. In particular, one can choose k to be the minimum natural number with such property.

Here, recognize the endomorphism $0 = f^k + c_1 b f^{k-1} + ... + c_{k-1} b f + c_k b$ is given by a multiplication map:

$$\begin{aligned} \forall m \in M, \quad \left(f^k + c_1 b f^{k-1} + ... c_{k-1} b f + c_k b\right)(m) &= \left(b^k + c_1 b^k + ... + c_{k-1} b^2 + c_k b\right) m \\ &= 0 \end{aligned} \tag{10.1}$$

Since M is flat (in particular, torsion-free), then all nonzero element $r \in R$ must have the multiplication map be injective (since R is an integral domain). Hence, the above multiplication map is 0 implies $b^k + c_1 b^k + ... c_{k-1} b^2 + c_k b = b\left(b^{k-1} + c_1 b^{k-1} + ... + c_{k-1} b + c_k\right) = 0$. With the assumption $b \neq 0$, one has $b^{k-1} + c_1 b^{k-1} + ... + c_{k-1} b + c_k = 0$. In particular, this implies the following:

$$b^{k-1} + c_1 b^{k-1} + ... + c_{k-1} b = -c_k \tag{10.2}$$

Which implies that $c_k \in (b) = I$.

Now, let's consider the endomorphism $f^{k-1} + c_1 b f^{k-2} + ... + c_{k-1} b + c_k$: Notice that it's monic (as polynomial of f), together with all non-leading coefficients $c_1 b, ..., c_{k-1} b + c_k \in I$. Also, any $m \in M$ satisfies the following:

$$\begin{aligned}\left(f^{k-1} + c_1 b f^{k-1} + ... + c_{k-1} b + c_k\right)m &= \left(b^{k-1} + c_1 b^{k-1} + ... + c_{k-1} b + c_k\right)m \\ &= 0 \cdot m = 0\end{aligned} \quad (10.3)$$

Which, $f^{k-1} + c_1 b f^{k-1} + ... + c_{k-1} b + c_k$ is a monic polynomial with coefficients in I, such that it evaluates to be 0 on M. This contradicts the assumption that k is the smallest natural number with such polynomial exists; as a result, one must have $N \cap Rx_n = 0$, showing $M = N \oplus Rx_n$.

Finally, since M is flat, then all of its direct summand must be flat. Hence, Rx_n is flat (implying $Rx_n \cong R$ with our base case), and $N = \sum_{i=1}^{n-1} Rx_i$ is flat. Which, since N only requires $\leq n-1$ generators, our induction hypothesis guarantees N to be free, or $N \cong \bigoplus_{i=1}^{l} R$ for some $l \in \mathbb{N}$.
This proves that $M \cong \bigoplus_{i=1}^{l+1} R$, which is a free R-module, and finishes our induction.

Hence, over R a PID, any finitely generated flat R-module M must be a free R-module.

11 D

Problem 11

Let R be a PID and let M be an R-module. Prove that M is a submodule of an injective R-module.

Solution: Recall that over R a PID, injective and divisible modules are equivalent. In particular, let K be the fraction field of R, then since K is divisible as R-module, hence it's injective.

Given the inclusion $R \hookrightarrow K$, any index set I generates an inclusion $\bigoplus_{i\in I} R \hookrightarrow \bigoplus_{i\in I} K$ by coordinate wise inclusion. In particular, notice that $\bigoplus_{i\in I} K$ is also divisible (since for any nonzero $r \in R$, any $(x_i)_{i\in I} \in \bigoplus_{i\in I} K$ has $r \cdot \left(\frac{x_i}{r}\right)_{i\in I} = \left(r \cdot \frac{x_i}{r}\right)_{i\in I} = (x_i)_{i\in I}$). So, any quotient of $\bigoplus_{i\in I} K$ (as R-module) is also divisible, which is injective.

Now, for any R-module M, we know there exists some index set I, such that the free R-module $\bigoplus_{i\in I} R$ has a surjection onto M (for the most extreme case, choose $I := M$, and associate each element of M with a copy of R). Hence, one has $M \cong \left(\bigoplus_{i\in I} R\right)/L$ for some submodule $L \hookrightarrow \bigoplus_{i\in I} R$.

Then, consider the composition of inclusions $L \hookrightarrow \bigoplus_{i\in I} R \hookrightarrow \bigoplus_{i\in I} K$, this realizes L as a submodule of $\bigoplus_{i\in I} K$. Hence, one can consider the module theoretic quotient $\left(\bigoplus_{i\in I} K\right)/L$. Which, this generates an R-linear map $\varphi : \bigoplus_{i\in I} R \hookrightarrow \bigoplus_{i\in I} K \twoheadrightarrow \left(\bigoplus_{i\in I} K\right)/L$. Notice that one has an element $(a_i)_{i\in I} \in \ker(\varphi)$ iff $(a_i)_{i\in I} \in L$ (since when including into $\bigoplus_{i\in I} K$, it must lie in L), showing that $\ker(\varphi) = L$. As a result, it factors uniquely to an injective R-linaer map $\overline{\varphi} : \left(\bigoplus_{i\in I} R\right)/L \hookrightarrow \left(\bigoplus_{i\in I} K\right)/L$:

$$\begin{tikzcd}
\bigoplus_{i\in I} R \arrow[rr, "\varphi"] \arrow[dr, two heads, "\pi"'] & & \left(\bigoplus_{i\in I} K\right)/L \\
& \left(\bigoplus_{i\in I} R\right)/L \arrow[ur, hook, "\overline{\varphi}"'] &
\end{tikzcd}$$

And, with $M \cong \left(\bigoplus_{i\in I} R\right)/L$, we have an inclusion of M into the injective R-module $\left(\bigoplus_{i\in I} K\right)/L$, finishing the desired claim.

12 D

Problem 12

Let R be a commutative ring and let m be a given maximal ideal of R. Suppose M is an R-module such that $M_m = 0$. Prove or disprove that $M = 0$.

Solution: We'll disprove the statement by providing a counterexample.

Consider the ring $R = \mathbb{Z}$, its maximal ideal $m = 2\mathbb{Z}$, and consider the $\mathbb{Z}$-module $M = \mathbb{Z}/3\mathbb{Z}$. It's clear that $M \neq 0$. However, $M_m = 0$, since one of the descripction of M_m is all element $\frac{m}{s}$, where $m \in M$ and $s \in \mathbb{Z} \setminus 2\mathbb{Z}$ (or, s is an odd number). Hence, recall that $3 \cdot m = 0$ for all $m \in M = \mathbb{Z}/3\mathbb{Z}$, then one has the following:

$$\forall m \in M, \quad s \in \mathbb{Z} \setminus 2\mathbb{Z}, \quad \frac{m}{s} = \frac{3 \cdot m}{3s} = \frac{0}{3s} = 0 \tag{12.1}$$

The reason is because $3 \in \mathbb{Z} \setminus 2\mathbb{Z}$, so the multiplication $3s \in \mathbb{Z} \setminus 2\mathbb{Z}$ makes sense. As a result, $M_m = 0$.

Hence, we have $m \subsetneq R$ is a maximal ideal, $M \neq 0$, but $M_m = 0$, which is a desired counterexample to the statement.

13 D (Referred to Problem 2)

Problem 13

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be two ring homomorphisms such that S is R-flat and T is S-flat. Prove or disprove that T is R-flat.

Solution: We'll prove that T is also R-flat, via the isomorphism proved in **Problem 2**, i.e. the isomorphism $(M \otimes_R N) \otimes_S P \cong M \otimes_R (N \otimes_S P)$, where $M \in \mathsf{R\text{-}Mod}$, $P \in \mathsf{S\text{-}Mod}$, and N an (R, S)-bimodule.

Let's first verify the (R, S)-bimodule structure on both S, T.
Given any $r \in R$, $s, s' \in S$ and $t \in T$, one has the following:

$$r \cdot (s \cdot s') = \varphi(r)(ss') = (\varphi(r)s)s' = (s\varphi(r))s' = s \cdot (\varphi(r)s') = s \cdot (r \cdot s') \tag{13.1}$$

$$r \cdot (s \cdot t) = \psi(\varphi(r))(\psi(s)t) = \psi(\varphi(r)s)t = \psi(s\varphi(r))t = \psi(s)(\psi(\varphi(r))t) = s \cdot (r \cdot t) \tag{13.2}$$

Since the actions of R, S commutes (if assuming they're commutative, under most cases we work with), these realize S, T as (R, S)-bimodules, hence the isomorphism can be used.

Now, we have the following isomorphism as both R and S-module:

$$T \xrightarrow{\sim} S \otimes_S T, \quad t \mapsto 1_S \otimes_S t \tag{13.3}$$

Also, for any inclusion of ideals $\iota : I \hookrightarrow R$, based on the R-flatness of S, we have $I \otimes_R S \hookrightarrow R \otimes_R S$ being injective. Then, when viewed this as an S-linear map, the S-flatness of T guarantees the map $(I \otimes_R S) \otimes_S T \hookrightarrow (R \otimes_R S) \otimes_S T$ being injective.

Finally, when consider the inclusion of ideals $\iota : I \hookrightarrow R$, it's tensor with T generates the following commutative diagram (based on all isomorphisms mentioned above):

$$\begin{tikzcd}
I \otimes_R T \arrow[r, "\iota \otimes \mathrm{id}_T"] \arrow[d, hook, two heads] & R \otimes_R T \\
I \otimes_R (S \otimes_S T) \arrow[r, "\iota \otimes \mathrm{id}_{S \otimes_S T}"] \arrow[d, hook, two heads] & R \otimes_R (S \otimes_S T) \arrow[u, hook, two heads] \\
(I \otimes_R S) \otimes_S T \arrow[r, hook, "(\iota \otimes_R \mathrm{id}_S) \otimes_S \mathrm{id}_T"] & (R \otimes_S S) \otimes_S T \arrow[u, hook, two heads]
\end{tikzcd}$$

As a result, since $I \otimes_R T \cong (I \otimes_R S) \otimes_S T$, $(R \otimes_S S) \otimes_S T \cong R \otimes_R T$, and the flatness conditions on S, T provides the injection $(I \otimes_R S) \otimes_S T \hookrightarrow (R \otimes_R S) \otimes_S T$, one has the map $\iota \otimes_R \mathrm{id}_T : I \otimes_R T \to R \otimes_R T$ also be injective.
With the inclusion $\iota : I \hookrightarrow R$ be arbitrary ideal of R, this shows that T is R-flat, proving the statement.

14 D

Problem 14

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R-flat. Prove or disprove that φ can not be surjective.

Solution: We'll disprove it by providing a counterexample.

Consider the rings $R = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z}$, $S = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$, and $T = \mathbb{Z}/2\mathbb{Z}$ (each with coordinate-wise addition and multiplication). For the homomorphisms, consider $\varphi : R \twoheadrightarrow S$ as the projection onto the first two coordinates $\varphi\left(\overline{a}_2, \overline{b}_3, \overline{c}_5\right) = \left(\overline{a}_2, \overline{b}_3\right)$, and $\psi : S \twoheadrightarrow T$ as the projection onto the first coordinate $\psi\left(\overline{a}_2, \overline{b}_3\right) = \overline{a}_2$.

Notice the composition $\psi \circ \varphi : R \twoheadrightarrow T$ is the projection onto the first factor $\psi \circ \varphi\left(\overline{a}_2, \overline{b}_3, \overline{c}_5\right) = \psi\left(\overline{a}_2, \overline{b}_3\right) = \overline{a}_2$. Meanwhile, if viewing T as an R-module, notice $\psi \circ \varphi$ as R-linear map has a section: Define $j : T \to R$ by $j(\overline{a}_2) = \left(\overline{a}_2, \overline{0}_3, \overline{0}_5\right)$. Let's verify it's R-linear:

$$\forall \overline{a}_2, \overline{b}_2 \in T, \quad j\left(\overline{a}_2 + \overline{b}_2\right) = \left(\overline{a_2} + \overline{b}_2, 0, 0\right) = (\overline{a}_2, 0, 0) + \left(\overline{b}_2, 0, 0\right) = j(\overline{a}_2) + j\left(\overline{b}_2\right) \tag{14.1}$$

$$\begin{aligned} \forall \left(\overline{a}_2, \overline{b}_3, \overline{c}_5\right) \in R, \ \ \overline{d}_2 \in T, \quad & j\left(\left(\overline{a}_2, \overline{b}_3, \overline{c}_5\right) \cdot \overline{d}_2\right) = j\left(\psi \circ \varphi\left(\overline{a}_2, \overline{b}_3, \overline{c}_5\right)\overline{d}_2\right) \\ & = j\left(\overline{a}_2 \overline{d}_2\right) = \left(\overline{a}_2 \overline{d}_2, 0, 0\right) = \left(\overline{a}_2, \overline{b}_3, \overline{c}_5\right)\left(\overline{d}_2, 0, 0\right) \\ & = \left(\overline{a}_2, \overline{b}_3, \overline{c}_5\right) j\left(\overline{d}_2\right) \end{aligned} \tag{14.2}$$

These verified j is R-linear. And, consider the composition $(\psi \circ \varphi) \circ j : T \to T$, we get:

$$\forall \overline{a}_2 \in T, \quad (\psi \circ \varphi) \circ j(\overline{a}_2) = (\psi \circ \varphi)(\overline{a}_2, 0, 0) = \overline{a}_2 \tag{14.3}$$

Hence, j is indeed a section of $\psi \circ \varphi : R \twoheadrightarrow T$, showing this surjection splits. As a consequence, it realizes T as a direct summand of R, hence T is projective as R-module, which is flat.

This is a counterexample to the statement, as T is R-flat under $\psi \circ \varphi : R \twoheadrightarrow T$, while $\varphi : R \twoheadrightarrow S$ is a projection, which is surjective.

15 D

Problem 15

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R-flat. Prove or disprove that ψ can not be surjective.

Solution: We'll disprove the statement by using the same counterexample in **Problem 14**.

In the previous problem we constructed $R = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z}$, $S = \mathbb{Z}/2\mathbb{Z} \oplus Z/3\mathbb{Z}$, and $T = \mathbb{Z}/2\mathbb{Z}$; the homomorphisms are projections $\varphi : R \twoheadrightarrow S$ and $\psi : S \twoheadrightarrow T$.

Then, T is projective as R-module (as it's a direct summand of R), hence R-flat. However, the second map $\psi : S \twoheadrightarrow T$ here is a projection, hence surjective. It's a desired counterexample to this statement also.

16 D

Problem 16

Let $R \xrightarrow{\varphi} S$ be ring homomorphism which is flat. Let M be an R-module such that $M \otimes_R S$ is torsion-free S-module. Is M torsion-free R-module?

Solution: We'll prove that M is not necessarily a torsion-free R-module, by constructing a counterexample.

Consider the ring $R := \mathbb{Z} \oplus \mathbb{Q}$ (with coordinate-wise addition and multiplication). Then, the projection onto the second coordinate $\pi_2 : R \twoheadrightarrow \mathbb{Q}$ defines $\mathbb{Q}$ as an R-algebra. Now, notice that as R-linear map, π_2 has a section, given by $j : \mathbb{Q} \to R$ as $j(q) = (0, q)$. Let's verify it has R-linear structure:

$$\forall q, r \in \mathbb{Q}, \quad j(q + r) = (0, q + r) = (0, q) + (0, r) = j(q) + j(r) \tag{16.1}$$

$$\forall (n, p) \in R, \quad j((n, p) \cdot q)j((\pi_2(n, p))q) = j(pq) = (0, pq) = (n, p) \cdot (0, q) \tag{16.2}$$

Hence, j is indeed an R-linear map. And, notice that any $q \in \mathbb{Q}$ satisfies $\pi_2 \circ j(q) = \pi_2(0, q) = q$, showing the projection π_2 splits.

This realizes $\mathbb{Q}$ as a direct summand of R when viewed as an R-module. Hence, $\mathbb{Q}$ is a direct summand of a free R-module, which is a projective R-module, in particular is flat.

Afterward, we'll denote π_2 as φ, and $\mathbb{Q}$ as S, so $\varphi : R \to S$ defines S as a flat R-module.

Now, consider $M := \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Q}$ as a ring (with coordinate-wise addition and multiplication again), and another projection map $\psi : R \twoheadrightarrow M$ by $\psi(n, p) := (\overline{n}, p)$ (where $\overline{n} \in \mathbb{Z}/2\mathbb{Z}$ is the quotient of n). This again realizes M as an R-module.

Notice that M as an R-module is not torsion-free: Consider the element $(2, 1) \in R$, and $\left(\overline{1}, 0\right) \in M$. The former element is not a zero-divisor in R, as if any $(n, p) \in R = \mathbb{Z} \oplus \mathbb{Q}$ satisfies $(2, 1) \cdot (n, p) = (2n, p) = (0, 0)$, one requires $p = 0 \in \mathbb{Q}$ and $2n = 0 \in \mathbb{Z}$, which also implies $n = 0 \in \mathbb{Z}$; on the other hand, $\left(\overline{1}, 0\right) \in M$ is a nonzero element. However, notice the action of $(2, 1) \in R$ on $\left(\overline{1}, 0\right) \in M$ is as follow:

$$(2, 1) \cdot \left(\overline{1}, 0\right) = (\psi(2, 1))\left(\overline{1}, 0\right) = \left(\overline{2}, 1\right)\left(\overline{1}, 0\right) = \left(\overline{0}, 1\right)\left(\overline{1}, 0\right) = \left(\overline{0}, 0\right) \tag{16.3}$$

So, the multiplication by $(2, 1)$ (an element that's not zero-divisor) is not injective, showing M has torsion elements as R-module.

Finally, we ought to prove that $M \otimes_R S$ is a torsion-free S-module. For this, let's recall the map $\psi : R \twoheadrightarrow M$ given by $\psi(n, p) = (\overline{n}, p)$. Hence, one has $(n, p) \in \ker(\varphi)$ iff $\overline{n} = 0 \in \mathbb{Z}/2\mathbb{Z}$ and $p = 0 \in \mathbb{Q}$, which is equivalent to $n \in 2\mathbb{Z}$ and $p = 0 \in \mathbb{Q}$. Therefore, $\ker(\varphi) = \{(2n, 0) \in R \mid n \in \mathbb{Z}\}$.

Now, consider the following exact sequence of R-modules:

$$0 \longrightarrow \ker(\varphi) \xhookrightarrow{\iota} R \xrightarrow{\psi} M \longrightarrow 0$$

where ι is the inclusion of ideals. By the flatness of S as an R-module proven before, we get the following exact sequence of R-modules:

$$0 \longrightarrow \ker(\varphi) \otimes_R S \xhookrightarrow{\iota \otimes \mathrm{id}_S} R \otimes_R S \xrightarrow{\psi \otimes \mathrm{id}_S} M \otimes_R S \longrightarrow 0$$

Here, we claim that $\ker(\varphi) \otimes_R S$ is 0: From above, its generators are of the form $(2n, 0) \otimes p$, where $(2n, 0) \in \ker(\varphi)$ (or $n \in \mathbb{Z}$) and $p \in S = \mathbb{Q}$. But, based on the way we define their R-module structure, we have the following:

$$\begin{aligned} (2n,0) \otimes p &= ((1,0)(2n,0)) \otimes (1p) = (1,0) \cdot_R ((2n,0) \otimes (\varphi(0,1)p)) \\ &= (1,0) \cdot_R ((2,0) \otimes ((0,1) \cdot_R p)) = (1,0) \cdot_R ((0,1) \cdot_R ((2n,0) \otimes p)) \\ &= ((1,0)(0,1)) \cdot_R ((2n,0) \otimes p) = (0,0) \cdot_R ((2n,0) \otimes p) = 0 \end{aligned} \tag{16.4}$$

(Note: To prevent confusion, here the R-action is denoted using $\cdot_R$, while the regular multiplication in each ring is without a dot).

Which, all generators of $\ker(\varphi) \otimes_R S$ is 0, showing that $\ker(\varphi) \otimes_R S = 0$. As a result, one has $R \otimes_R S \xrightarrow{\psi \otimes \mathrm{id}_S} M \otimes_R S$ being an isomorphism by the exactness, and as R-module one has $R \otimes_R S \cong S$, so $M \otimes_R S \cong S$ as R-module.

Then, when realized as S-module, since $S = \mathbb{Q}$, then $M \otimes_R S \cong S$ is in fact a 1-dimensional S -vector space, which is torsion free as S-module.

Hence, the above example shows a ring homomorphism $R \xrightarrow{\varphi} S$ which is flat, M an R-module such that $M \otimes_R S$ is torsion-free as an S-module, yet M itself is not torsion-free as R-module. This provides a counterexample to the statement.